

Problem Solving with Parallelograms and Rectangles

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How to Solve Parallelograms and Rectangles

Step 1: Identify the Figure

Identify whether the figure is a **parallelogram** or a **rectangle**.

Step 2: Identify the Property

Determine the property that relates the given and unknown measures.

Properties of Parallelograms

In a **parallelogram**:

Opposite sides are congruent.

Opposite angles are congruent.

Consecutive angles are supplementary.

Diagonals bisect each other.

Properties of Rectangles

In a **rectangle**:

Opposite sides are congruent and parallel.

All angles are right angles.

Diagonals are congruent and bisect each other.

Step 3: Write an Equation

Write an equation using the appropriate property.

For example, if opposite sides of a parallelogram are congruent,

$$AB = CD$$

Step 4: Solve

Solve the equation, then substitute the value to find the required side length, angle measure, or diagonal length.

Step 5: Check

Check that your answer satisfies the given conditions.

Examples

Example 1

In parallelogram $ABCD$, $AB = 3x + 5$ and $CD = 20$. Find x .

Opposite sides of a parallelogram are congruent.

$$3x + 5 = 20$$

$$3x = 15$$

$$x = 5$$

Example 2

In parallelogram $JKLM$, $\angle J = 2x + 15^\circ$ and $\angle K = 5x - 9^\circ$. Find x .

Consecutive angles are supplementary.

$$(2x + 15) + (5x - 9) = 180$$

$$7x + 6 = 180$$

$$7x = 174$$

$$x = \frac{174}{7}$$

Example 3

In parallelogram $WXYZ$, $WX = 4x - 7$ and $YZ = 3x + 5$. Find x .

Opposite sides are congruent.

$$4x - 7 = 3x + 5$$

$$x = 12$$

Example 4

In rectangle $EFGH$, the length is 15 cm and the width is 8 cm. Find the area.

$$A = lw$$

$$A = (15)(8)$$

$$A = 120 \text{ cm}^2$$

Example 5

The length of a rectangle is $(2x + 4)$ cm and its width is $(x - 1)$ cm. If its perimeter is 34 cm, find x .

$$P = 2(l + w)$$

$$34 = 2[(2x + 4) + (x - 1)]$$

$$17 = 3x + 3$$

$$x = \frac{14}{3}$$

Practice Exercises

Practice 1

In parallelogram $ABCD$, $AB = 3x + 5$ and $CD = 20$. Find x .

Practice 2

In parallelogram $JKLM$, $\angle J = 2x + 15^\circ$ and $\angle K = 5x - 9^\circ$. Find x .

Practice 3

In parallelogram $WXYZ$, $WX = 4x - 7$ and $YZ = 3x + 5$. Find x .

Practice 4

In rectangle $EFGH$, the length is 15 cm and the width is 8 cm. Find the area.

Practice 5

The length of a rectangle is $(2x + 4)$ cm and its width is $(x - 1)$ cm. If its perimeter is 34 cm, find x .

Activities

Activity 1

In parallelogram $QRST$, $\angle Q = 3x + 25^\circ$ and $\angle S = 5x - 15^\circ$. Find x and the measure of $\angle Q$.

Activity 2

The diagonals of parallelogram $ABCD$ intersect at E . If $AE = 3x - 1$ and $EC = 20$, find x .

Activity 3

In rectangle $PQRS$, the length is 22 cm and the width is 9 cm. Find the perimeter.

Activity 4

In rectangle $ABCD$, diagonal $AC = 4x + 6$ and diagonal $BD = 2x + 18$. Find x .

Activity 5

The area of a rectangle is 108 cm^2 . If its width is 9 cm , find its length.

Thank you!